

CSE260 Assignment - 2

Deadline - 20th July, 11:59 PM

Marks: 25

Assignment must be handwritten. Scan and upload PDF in the given google form

1. Simplify the following Boolean expression [5]

$$xyz' + x'yz + xyz + x'yz'$$

2. Find the complement of the expression [5]

$$(a'b + cd)e' + e$$

3. Find out SOP and POS for the following [You cannot use truth table to solve these]: [5+5 = 10]

$$F(V,W,X,Y,Z) = WY + WX + X'Y$$

4. Draw the following functions using NAND gates only. Use a maximum of 2 inputs for each gate [5]

$$F(A,B,C,D) = (ABCD + A'D' + (B'+D)')'$$

Ungraded Problems (Only for practice, Not required to submit with assignment):

1. Simplify the following boolean functions using laws of Boolean algebra and draw the simplified function using NAND gates only.

$$F(a,b,c,d) = \sum(2,5,7,10,11,13)$$

2. Draw the following functions using NOR gates only:

$$F(A,B,C,D) = (ABCD + A'D' + (B'+D)')$$